

Chapter 17

International Research Collaborations Between Universities and Industry



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17.1 Introduction

Universities and industry can mutually benefit from joint research collaborations in various ways. Universities can offer industry-unique research assets, such as specialized access to their laboratories, professors, and students. They can also enable industry to leverage federal funding for joint research in areas of common interest. Likewise, industry brings practical perspectives and real-world experiences into conversations, stimulating new research ideas and opening additional funding opportunities.

Highlight—International Collaborations

International University-Industry (UI) collaborations are crucial in addressing global challenges and driving innovation. These partnerships combine universities' research resources and expertise with the technical capability and funding from industry to accelerate scientific discovery and innovation. This chapter explores the dynamics of international UI collaborations, focusing on unique opportunities and challenges such as aligning goals, managing intellectual property, and navigating geopolitical and cultural barriers. Successful

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international UI collaborations require time, clear communication, trust, and shared values. Case studies from partnerships like Mars and the University of California, Davis, highlight the benefits of international research collaboration in food sustainability, climate-smart agriculture, and health. The chapter also highlights recommendations for improving international collaboration success through strategic alignment of goals, efficient resource use, and strong management frameworks.

Academia is designed for and specializes in basic research, in which fundamental scientific understanding is prioritized. On the other hand, industry is built for and focuses on applied research, which focuses on solving real-world problems. By coming together, use-inspired efforts can be effectively conducted, leading to breakthrough capabilities. This is best expressed in Stokes’ model of scientific research, which divides scientific research into four quadrants. Pasteur’s Quadrant, signifying use-inspired basic research, is the model’s section that applies to international research collaborations. In Pasteur’s Quadrant, academic and industry efforts merge, involving basic research motivated by practical, real-world applications (Stokes, 1997) (Fig. 17.1).

Research collaborations, where scientists combine their skills, knowledge, and research for a common goal, are fundamental science activities (Bozeman & Boardman, 2014). International research collaborations, a subset of traditional research collaborations between academia and industry, occur when researchers work across national borders (Gok & Karaulova, 2024). Over the past two decades,

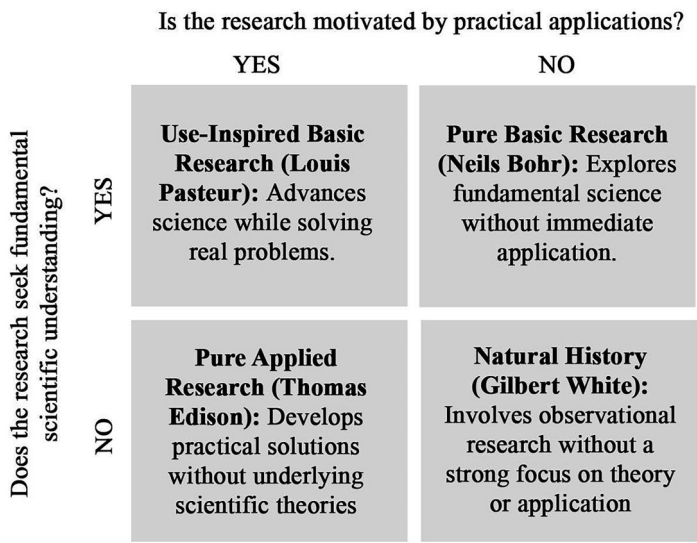


Fig. 17.1 Quadrant model of scientific research (Adapted from Stokes, 1997)

there has been a noticeable increase in published research from international collaborations, particularly those involving three or more countries (Maher & Van Noorden, 2021).

International research teams have unique requirements for success and often encounter specific challenges and opportunities. This chapter analyzes current research on this topic and provides specific examples of successful international collaborations. Several recommendations are also included to increase the impact of international research and translational efforts, given that we operate in an increasingly globalized society.

International research collaborations can take various forms. Collaborations between universities and industries can involve basic research on a particular topic or applied research for a specific industry challenge. Industry can sponsor such basic or applied research, or the funding can come from universities, government, or nonprofit grant programs. Technology transfer, the process of transferring technology from a university to a business, is sometimes the goal or the result of the collaboration. Once transferred, the technology can be developed into a commercial product or a service (U.S. Patent and Trademark Office, [n.d.](#)). A company may also sponsor postdoctoral researchers to help build technical capability in specific scientific areas. Similarly, industry partners may choose to fund university research centers on campus.

17.2 Opportunities and Challenges

Working within an international or multinational team can benefit both universities and industry partners. For instance, international research collaborations can broaden knowledge, facilitate the development of theoretical approaches and methods, and help distribute the labor required within the research (Dusdal & Powell, 2021). Furthermore, publications associated with international coauthors have been found to have a greater research impact (Levitt & Thelwall, 2010).

Researchers often tackle global challenges that are complex in scope and scale and are typically very difficult to solve alone. They often benefit from bringing together diverse perspectives, experiences, and technical areas of expertise apparent in international collaborations (UIDP, 2022). Both academia and industry can combine their strengths, increasing the possibility of accelerating innovation for real-world impact (UIDP, 2022).

Such international research collaborations can drive value for both parties, but several challenges could be considered before beginning a partnership. We outline these considerations below under these categories: (1) Strategy and Alignment of Goals, (2) Know-how, (3) Intellectual Property (IP) Rights, (4) Geopolitical Considerations, and (5) Culture, Communications, and Logistics.

Many of these considerations are unique to international research collaborations (vs. those between parties in a single country). For instance, language barriers may arise between collaborators from different countries, resulting in misunderstandings

(Yao, 2021). Similarly, cultural differences relating to communication styles may create challenges. International research collaborators should also be aware of the policies and laws of the federal government or funding agency sponsoring the research; consulting knowledgeable legal experts can help ensure understanding of these policies before research begins (Yao, 2021).

17.2.1 Strategy/Alignment of Goals

Potential university and industry partners need to share similar goals involving the research subject matter. One success factor in creating a new research partnership is a shared interest in the research topic (Elsevier, 2021). This might seem simple; however, it often takes time and direct communication to ensure such alignment. This misalignment of university researchers and industry research managers' goals is a common barrier to university–industry (UI) partnerships (Association of Public and Land-Grant Universities, APLU, 2022). The cause of misalignment can be due “more to perception than reality” and can be alleviated by increased communication and clarification about how joint goals and incentives can be supported (APLU, 2022).

Aligning common research interests may not be enough to ensure collaborative success. Aligning incentives can be even more important. If incentive structures are such that both parties benefit from solving a common problem, the collaboration is mutually beneficial and more likely to succeed.

In addition to incentives to achieve common goals, it is crucial to articulate mutual value. In interviews with industry and university representatives, APLU (2022) found that universities seek partners with shared interests in scientific discovery that produces value for the industry. However, while interested in scientific discovery and resolving grand challenges to solve social problems, industry stakeholders tend to assess joint research activities with universities on the return on investment (APLU, 2022). Academia is typically incentivized for their research (e.g., publications and preliminary data for future research endeavors). Articulating the potential to deliver value for both parties and having a shared understanding of that value is crucial.

17.2.2 Know-How

Universities and industry might share a desire to take the first step toward engaging internationally but need more experience to initiate such relationships. This could be particularly true in cases where limited experience with international research collaborations is required to address a common challenge or a lack of current collaboration with international parties. Companies and academic institutions have regional, national, and international connections within their networks. It is wise to create a map of one's current network as a starting point to understand the global

touchpoints available within one's institution. To complement this, various tools are available for further research.

To help find technical experts in a field of interest, industry associates can use online platforms, like Halo, created by Halo Science. Halo is an online, AI-powered platform that helps R&D teams expand their current networks and link them with international technical expertise for their bespoke research needs. This can help the industry avoid blind spots (by only collaborating with their limited, known networks) and commercialize innovations more quickly (Halo Science, [n.d.](#)). Likewise, university researchers can meet new industry partners across the globe in their fields of technical interest by creating an account on Halo.

University and industry associates may also access Elsevier's six-module tool, SciVal. This platform allows users to access the performance data of 24,000 research institutions in over 30 countries (Elsevier, [n.d.](#)). SciVal allows analysis of collaboration partners in its Benchmarking module and the identification of potential industry and academic partners for global collaboration partnerships in its Collaboration module. Users can gain insights about technical expertise across sectors and research areas of strategic value.

Alternatively, researchers can access web-based tools to find country-based government awards and funded research opportunities. For instance, the European Commission has a robust online tool called the EU Funding and Tenders Portal, which links researchers to projects and programs (European Commission, [n.d.](#)). One can find calls for proposals, opportunities to work as a technical expert, and projects previously funded and their results. Importantly, there is also a feature to search for researchers and organizations currently involved in EU-organized programs, such as Horizon Europe (UIDP, [2024](#)). The United States also offers platforms to research grant funding and programming via the National Science Foundation (NSF), the National Institutes of Health (NIH), and the US Department of Energy (DOE), among others.

Though these web-based tools can allow researchers to explore global collaboration opportunities, the relationships needed for success do take time. Carefully balancing open innovation opportunities and ensuring trusted partnerships among collaborators is needed.

17.2.3 Intellectual Property Rights

IP is at the crux of many international research collaborations. Patents on discoveries or designs, copyrights on data models, and trade secrets on research methodologies are all possible outputs of joint research (Iqbal, [2024](#)). It is important to note that there are no "international" patents, and laws are not harmonized. However, there have been several initiatives over the years to build frameworks to reduce inefficiencies and redundancies in the application process (National Academies of Science, [2011](#)). Success in IP negotiations can be supported by understanding the differences in perspectives between industry and academia (including public vs.

private universities and small vs. large companies) (National Academies of Science, 2011).

Transparent IP rights and policies benefit university researchers and industry partners by enabling both parties to engage suitably in a way that will benefit them in the long term (Dollinger et al., 2018). This is supported by the findings of Bruneel et al. (2010), who found that the industry may have reservations about confidentiality and IP rights, further highlighting the need for all terms to be aligned and agreed upon before partnering.

Securing pre-negotiated rights is essential to ensure the long-term success of relationships throughout the licensing process. This is particularly important, as a “first right to negotiate” can create uncertainty for companies, leaving them unsure of when or how they will gain access to the asset they are investing in. Furthermore, negotiations tend to become more competitive when the asset is valuable, as both parties push harder for favorable terms. In contrast, when the asset is still in its conceptual stage, discussions typically focus more on market standards and fairness than the asset’s concrete value. It is also crucial to accurately define a market standard range for expected inventions or assets prior to collaborating.

Ensuring compliance within global IP frameworks is essential. The capability to protect IP rights across international borders must be in place, either through in-house legal talent or external consultancies. The World Intellectual Property Organization (WIPO) portal is an excellent place to start, as one can find IP regulations by country, information on international patents and trademarks, and IP dispute resolution (WIPO, n.d.).

Companies, particularly those based outside the United States, should be aware that the US government mandates that land-grant universities maintain ownership of their patents via the 1980 Bayh–Dole Act (previously, patents funded by government research were owned by the government). This model may be challenging to navigate for companies accustomed to a different IP approach.

17.2.4 Geopolitical Considerations

A geopolitical perspective is vital when beginning new international collaborations and nurturing existing ones. Some have found that political divergence can negatively affect the development and preservation of cross-border research collaborations (Charpin et al., 2024). In cases of rising geopolitical tensions, it can be challenging to operate freely; therefore, it is recommended that partners reflect on changing geopolitical landscapes as they develop international collaborations (Charpin et al., 2024). There is evidence of recent research slowdowns between some countries that may be rooted in geopolitical tensions, leading to increased scrutiny of international researchers (Maher & Van Noorden, 2021). Also affecting the breadth of global research is the reduction of visas for visiting scholars and university students (Maher & Van Noorden, 2021).

Though much of the research at universities may not be directly impacted by geopolitical relations, some universities are cutting back on global research engagements due to geopolitical considerations. A survey of over 700 universities in over 100 countries has shown decreasing participation both globally and regionally, with geopolitical tensions decreasing global research links in Europe and North America more so than in other areas of the world (Latin America, Caribbean, Africa, and the Middle East) (Matthews, 2024). 11% of respondents have revised research partnerships due to newly introduced government regulations, and 3% have suspended research partnerships with certain countries by choice (Marinoni & Cardona, 2024).

The concept of sovereignty can arise when experts from different countries collaborate on scientific research. Countries have the authority to control biological resources within their borders. In 2014, the Nagoya Protocol was negotiated to promote the fair and equitable sharing of benefits from genetic resources (Covington, 2024). In practice, the Protocol would arise when those collecting genetic resources share their benefits with the countries where the specimens originated. This would directly benefit countries with fewer resources. The complexities of the Nagoya Protocol are found in the nuances of its implementation. Each country can have its own legislation and best practices, and codes of conduct are not standardized (Watanabe, 2015). Various administrative complications can arise, like the need for permits to use genetic resources from their own country; if the proper permits are not obtained, subsequent sequencing can be illegal in the country where the genetic resources are being tested (Watanabe, 2015). Knowing and understanding the Nagoya Protocol and other international legal policies is crucial before undertaking international research. Consulting with legal experts knowledgeable in international research policies is recommended.

17.2.5 Culture, Communication, and Logistics

Teamwork can be challenging, including within international UI collaborations. Specific requirements are needed for the best chance of success. As its name implies, international research collaborations often involve researchers across time zones and continents. Trust, support, and friendship are crucial to success, particularly beyond the project. For example, the drafting of publications and the design of follow-up projects can benefit from fellow researchers' mutual trust and support (Dusdal & Powell, 2021). Particularly in the early phases of a relationship, academic and industry researchers must devote time to learning and appreciating the culture and norms of their likely disparate organizations; this may include reconciling differences to build trust and common understandings (Bruneel et al., 2010). Experience can help overcome collaborative barriers. If groups begin with smaller projects, they can gradually increase complexity, benefiting both parties and prolonging the partnership (Rossoni et al., 2023).

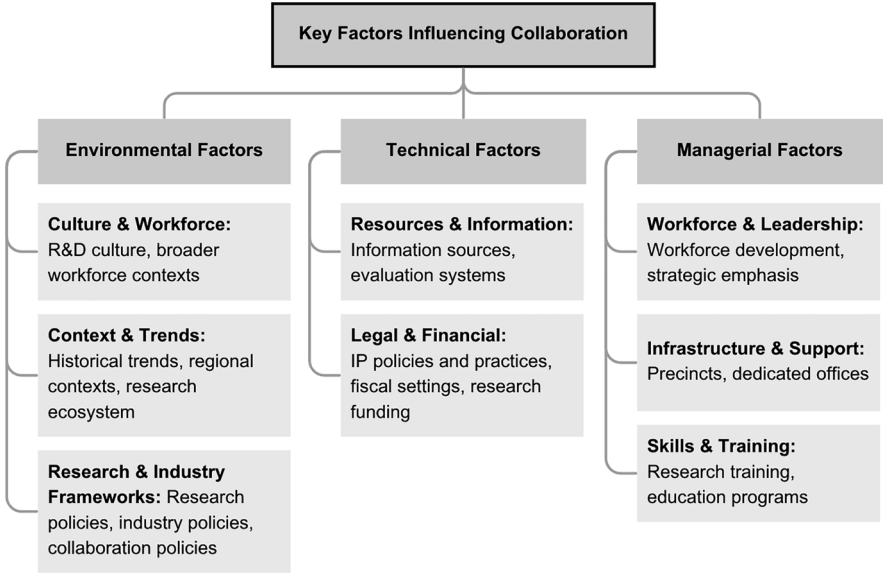


Fig. 17.2 Framework for international UI collaboration (Dollinger et al., 2018)

17.2.6 Historical Frameworks

Several frameworks explain international university and industry research collaborations. Dollinger et al. (2018) provide three factors to approach UI collaboration: environmental, technical, and managerial. These factors each have specific attributes. Figure 17.2 illustrates the framework, showing the factors and categorizing attributes, such as R&D culture and broader workforce context, which are categorized as culture and workforce. Progress against the three factors can help evaluate and benchmark UI collaborations.

Rossoni et al. (2023) evaluated barriers to UI research, development, and innovation across three theoretical perspectives. The first is the triple helix model of interactions between universities, industry, and governments for economic development and innovation. The second perspective focused on relational social capital and value creation, delivering ways to create value and facilitate collaborations. The third encompasses technology transfer and cultural differences.

17.3 Options, Solutions, and Impact

Industry and university groups have options to help sustain international research collaborations to drive innovation. Industry can consider co-location on university campuses to leverage academic resources and the technical expertise of faculty and

students while providing in-kind and financial support to the facility. Global consortia can be developed to tackle common goals and drive progress in technical areas.

17.3.1 Global Industry Co-location on University Campuses

While many industry scientists are located within focused R&D teams on-site at their respective companies, industry, and academia may choose to partner globally on research by co-locating their employees. UIDP defines co-location as “the purposeful combination of industry and university personnel in a dedicated space in which costs are shared for active collaborative or independent research with the strategic intent of encouraging idea exchange by reducing communication and cultural barriers that accompany the physical challenge of being located in different facilities” (UIDP, 2019). Co-location can offer many attributes by design, including joint events and research, two-way exchanges of ideas, access to shared equipment, leadership involvement, increased probability of joint publications and IP, and the opportunity for students to work hand in hand with technical industry leaders (UIDP, 2019).

Perhaps most indicative of this type of relationship is the one between Unilever and Wageningen University. Unilever, headquartered in London, UK, is a large consumer goods company. In 2024, over 128,000 people were employed worldwide by this multinational company, which has a vast network of over 280 factories and 57,000 suppliers (Unilever, [n.d.-a](#)). According to its corporate website, Unilever spent €949 million on R&D in 2023 and held a portfolio of over 20,000 patents and 5000 R&D experts in 60 countries (Unilever, [n.d.-b](#)). 500 employees were physically based at Wageningen University & Research (WUR) in Wageningen, Netherlands, at Unilever’s global Food Innovation Centre, called Hive.

Unilever began construction on Hive in 2016, and the space opened in 2019. Hive contains 18,000 m² of collaboration spaces, including a pilot plant, laboratories, and kitchens (Unilever, [n.d.-c](#)). Research at the Hive has included research with both organizations as well as research including additional parties. For instance, in 2024, Unilever reported a new 4-year research project with both WUR and IFF, a global leader in food, beverage, scent, home and personal care, and health based in New York City, USA (Unilever, 2024; IFF, 2024). This research focuses on improving plant-based meat flavor and texture research by exploring protein and flavor interactions.

Unilever was not the first company to partner with WUR on campus. In fact, in 2010, FrieslandCampina, the global dairy company headquartered in the Netherlands, concentrated its R&D teams at WUR. The Innovation Centre has a pilot plant with rooms for cooling and storage, laboratories, offices, an experience center, an innovation kitchen, and a bakery (FrieslandCampina, [n.d.](#)).

Though not technically co-located, Germany-based Bayer has an arrangement that physically brings them closer to their research collaborators. For over 10 years, Bayer, the Broad Institute of MIT, and Harvard University have collaborated

extensively to develop new oncology therapies (Bayer, 2023). The Bayer Research and Innovation Center (BRIC) is conveniently located in Cambridge, very close to the Broad Institute.

Another example of this model is the long-standing partnership between Mars and the University of California, Davis (UC Davis), which spans over four decades of international research collaboration. Mars, a large multinational company headquartered in the United States, embedded a technical leader on the UC Davis campus beginning in 2004, who became involved with many scientific research projects. During this period, Mars and UC Davis completed many major project reviews, established the first Research Advisory Board (focusing on plant sciences, specifically cacao, peanuts, and mint), and started the Multi-Disciplinary Research Unit, which brought Mars and UC Davis into greater global collaborations to answer questions raised by the R&D Community.

Mars established the Mars Advanced Research Institute (MARI) in 2013 as a corporate functional area with an operational hub located at UC Davis. When MARI was first formed, Mars had two key technical roles based on the campus. This created collaboration and technical knowledge exchange opportunities for US-based and international associates at both organizations. The unique organizational design of MARI enabled rapid and flexible access to a global network of scientific experts and infrastructure vital for the company's transformative innovation goals.

With the support of Mars, UC Davis constructed a greenhouse complex on campus. This complex allows UC Davis researchers, students, and Mars scientists to partner to develop climate-smart cacao. The collaboration has helped build climate resilience in cacao by examining how the cacao plants respond to various environmental stimuli. An extensive and diverse collection of cacao germplasm is maintained in the greenhouse, which is used to characterize the phenotypic traits and breed new clones with increased resistance to diseases and higher productivity, resulting in more cocoa beans per tree for producing quality chocolate (UC Davis, 2019). The greenhouse is part of a broad ecosystem at UC Davis, working on early-stage and applied research to help “futureproof” cocoa farming through science to enable climate-smart, resilient, and high-yielding cocoa trees. The greenhouse is also a starting point for new cacao clones for Mars, as the clones can then be trialed in the company's regional field research stations and ultimately deployed to cacao farmers.

17.4 Case Studies of International Research Collaboration Involving Mars and UC Davis

Mars and UC Davis have collaborated for decades across various scientific research fields, including over 100 projects in the last 10 years. Several of these initiatives are exemplary case studies highlighting this long-standing international partnership's tangible benefits and outcomes. Notably, their joint efforts have advanced research

in areas such as the alternate wetting and drying of rice, the African Orphan Crops Consortium, and the Innovation Institute for Food and Health.

Alternate Wetting and Drying (AWD) of Rice Rice is an important food staple globally for over half of the world's population (Fukagawa & Ziska, 2019). Traditional rice production brings sustainability challenges due to irrigation water requirements and methane emissions. The Mars plant sciences lead at the time had access to satellite data as a part of the California Nitrogen Assessment that showed clouds of nitrous oxides over Northern California rice paddies. Conversations were held between UC Davis, the California Rice Commission, and Mars about this finding. Mars proposed to conduct a series of field experiments to ameliorate N₂O emissions. Fundamental research was required to understand more about the measures that could be taken to improve the sustainability of rice growing.

Mars and UC Davis (and, later, the University of Arkansas) partnered to study a specific rice irrigation technique, first developed by the International Rice Research Institute (IRRI) called “alternate wetting and drying,” or AWD. AWD involves flooding a rice field and allowing the water to naturally subside to the soil's surface during the safe rice growth stage (Henry et al., 2017). This climate-smart agricultural method significantly reduces water use and greenhouse gas emissions and maintains yields when implemented properly. AWD can also be combined with other conservation practices to reduce irrigation water and energy consumption, improving farmers' profits and quality of life.

This work built upon the research UC Davis had been focusing on, driving forward the long-term sustainability of rice systems. A collaboration was established to study, adapt, and document the benefits of AWD in California first and then the Mississippi Delta. Mars and UC Davis conducted and published research about AWD with the University of Arkansas between 2011 and 2016 (Nalley et al., 2015). The early-stage promising results secured a 2018 grant by the National Resources Conservation Service of the U.S. Department of Agriculture (USDA), which awarded a grant to the White River Irrigation District in Arkansas to implement similar water management schemes on approximately 10% of the rice area. Farmers received financial support for adopting practices in this program, including limiting fertilizer nitrogen applications and reducing irrigation water use through AWD.

In the following years, the work on AWD in North America was complemented by similar efforts for European rice production areas. UC Davis helped transfer methodology to measure emissions in situ in rice fields to Mars partners in European rice research. Mars continued to introduce climate-smart agricultural practices, including broader adoption of AWD, to farmers within its supply chain in Asia.

This partnership was critical for Mars as it helped secure increased, high-quality rice yields. It also offered strategic benefits. For instance, Mars gained privileged access to a trusted network in the scientific and agricultural communities that complemented its distribution and manufacturing strengths. This collaboration, rooted in UC Davis' mission of transformative education, innovative research, and public service—and embodying Mars' principle of mutuality—drives breakthrough innovation and sustainable growth for both organizations. Engaging with premier

research institutions reinforces Mars' reputation as a leader in responsible, forward-thinking science. Grounded in mutuality and responsibility, this collaboration deepens stakeholder trust and demonstrates shared dedication to advancing research that meets commercial imperatives and societal needs. Mars associates also benefited from the direct interaction with top academics and exposure to cutting-edge research. Anchored in the Mars principle of mutuality, this environment of continuous learning equips teams with the insights and skills needed to drive global innovation.

The program also benefited the universities involved. Students were provided with immersive, experiential learning opportunities that bridge theory and practice. By cultivating technical and soft skills through a relationship founded on mutuality, UC Davis furthers its mission of transformative education and prepares graduates to excel in a globally interconnected marketplace. The partnership helped to elevate UC Davis' role as a hub for innovation and public service. This reciprocal relationship facilitates the exchange of groundbreaking research and innovative insights with regional, national, and global stakeholders, reinforcing our commitment to mutual growth and the common good. Direct engagement with Mars gives UC Davis unique insights into evolving market needs. This two-way dialogue enriches academic research and ensures that scholarly work remains relevant, socially responsible, and oriented toward addressing real-world challenges—further solidifying UC Davis' role as a catalyst for sustainable progress.

Companies may lack the infrastructure (e.g., labs or agricultural spaces) to research scientific topics of interest. In this case, the University of Arkansas had the needed agricultural infrastructure for the research on AWD, providing opportunities to test, verify, and learn. Universities can provide a suitable setting for companies to explore new ambitions, fine-tune strategies, and stimulate innovation.

The results of the AWD research included robust, detailed, and scientifically supported data. Data is integral for companies to support internal and external stakeholders to drive change, and rigor is crucial. Data can provide the backbone for scientific publications. Both universities and industry can benefit from robust data resulting from collaborations.

Maintaining continuity can be challenging in international research collaborations like this one involving AWD. Lack of continued funding can stymie future work, as universities cannot fund research independently. Shifts in leadership and staffing may lead to the reassessment of projects. It is essential to consider these factors when pursuing international research partnerships to ensure long-term success and stability.

African Orphan Crops Consortium (AOCC) Mars, The African Union—NEPAD, WWF, and the University of California, Davis established the African Orphan Crops Consortium in 2011 with an ambitious goal to end chronic hunger and malnutrition and eradicate stunting on the continent of Africa through year-around access to locally available nutritious food crops to break the intergenerational cycle of nutrient deficiency. The AOCC aims to improve the backbone of the African food system with a focus on nutrition, production, climate change resil-

ience, pest and disease resistance, and water and nutrient use efficiency. “Orphan” crops are little improved by modern plant science. These crops are dietary staples and provide income for the poorest African farmers. (Food Insight, 2015). The AOCC is an uncommon collaboration of over 30 government organizations, scientific and agricultural bodies, universities, companies, regional organizations, nongovernmental organizations, and a network of more than 15 agricultural and horticultural organizations. AOCC identified 101 traditional pan-African food crops for improvement. To date, whole-sequence data has been generated for 75 crops, 46 of which have been published (African Orphan Crops Consortium, 2023).

The African Plant Breeding Academy (AfPBA), coordinated by UC Davis, has trained more than 172 African scientists to do this work. AfPBA alumni have landed more than \$175 million in competitive grants, working on 143 improved crop varieties, including more than 40 AOCC-designated orphan crops. They have also published over 665 peer-reviewed scientific papers and established the African Plant Breeders Association (APBA). AfPBA alumni have emerged as leaders in plant breeding education, developing 62 graduate-level plant breeding courses based on knowledge and tools gleaned in AfPBA. These have impacted hundreds of young people, the next generation of crop improvement scientists. The results of the scientific work are made freely available to scientists on the Internet. Approved by African Heads of State at the African Union Assembly, the AOCC and its academy comprise the continent’s most comprehensive and integrated crop improvement venture, providing students and graduates with the best scientific instruments. The AOCC and the African Plant Breeding Academy are Africa’s most successful crop breeding training programs needed to deliver crop improvement and free the continent of chronic hunger and malnutrition.

Innovation Institute for Food and Health (IIFH) In 2015, Mars and UC Davis established the Innovation Institute for Food and Health (IIFH), focusing on food, agriculture, and health. The IIFH catalyzes academic-industry innovation with a vision to “make the food we love healthier for all people and the planet” (IIFH, n.d.). The IIFH now has a four-pronged approach, encompassing the development of future innovator talent in food and health, using new technologies to innovate healthier foods, enabling uncommon collaborations, and supporting new products and companies (IIFH, n.d.). The partnership included a financial commitment from Mars, matched partially by UC Davis, to develop and fund innovations from preliminary research to commercialization. In the 10 years since its inception, the IIFH has included 21 additional technical and investment partners (IIFH, n.d.).

Given the remit of the IIFH, many projects are global. For instance, the IIFH supported the Periodic Table of Food Initiative (PTFI), launched in 2022. PTFI is a global effort to standardize food analysis to understand food’s impact on human health, agriculture, and nutrition (The Rockefeller Foundation, 2022). PTFI aims to characterize and quantify biochemicals in food using advanced scientific techniques such as analytical chemistry, data processing, bioinformatics, and machine learning (The Rockefeller Foundation, 2022). By understanding the molecular composition

of food, the PTFI aims to share precise recommendations and formulations that create more nutritious and sustainable foods, thereby improving food quality and rediscovering the value in foods widely forgotten. The PTFI has a global network of partners across regions and sectors, including a “Center of Excellence” on each continent, including universities and public health institutions (Periodic Table of Food Initiative, 2024). The Centers of Excellence lead food quality research, capacity strengthening, and translation of science for impact within their regions.

Mars and the IIFH have collaborated on research to identify a new blue color for food coloring. Despite a long history of exploration, blue remains one of the most challenging colorants to obtain from any source, even more so from natural, edible sources. Blue is critical as it is necessary to produce other colors across the palette. Achieving a cyan blue from natural sources that could replace synthetic dyes has been a long-standing challenge to the food industry. Solving the challenge can enable a broader color palette for foods. Computational simulations and large-array spectroscopic techniques were used to determine this previously uncharacterized cyan blue anthocyanin-based colorant’s 3D chemical structure, color expression, and stability. Synthetic biology and computational protein design tools were leveraged to develop an enzymatic transformation of red cabbage anthocyanins into the desired anthocyanin. The new cyan blue, obtained from red cabbage, could be an alternative to synthetic blue food colorings. The research collaboration published a peer-reviewed manuscript on the findings, filed two patents, and co-founded Peak B, a startup company out of UC Davis, to continue development for scaling production, obtaining safety data, and defining product specifications to create a commercially viable product (Denish et al., 2021).

17.5 Takeaways

International collaboration on technical research offers numerous advantages for university and industry partners. Successful international research collaborations require patience and perseverance. It takes time to establish trust and working methods within the partnership. These partnerships enable tackling global challenges while benefiting from diverse perspectives, expanded networks, and enhanced access to resources and funding. However, such collaborations also present challenges, including geopolitical complexities and potential intellectual property disputes. Additionally, they demand careful attention to ensure alignment of strategies and goals and establish clear communication and logistical frameworks.

Authors’ Note and Acknowledgements

Mars and the University of California, Davis, have a long history of collaborating internationally over the last 40 years. Their relationship was formalized 10 years ago, creating a strategic framework of combined capabilities to help build sustainable food systems, find new solutions to increase health and wellbeing, and improve food safety. The international research collaboration between Mars and the

University of California, Davis, is grounded in scientific research across diverse fields of study, including agriculture, food science, nutrition, biology, and veterinary health (Fell, 2014). Critical joint international research collaborations have included sequencing the *Theobroma cacao* genome and founding the African Orphan Crops Consortium. Both organizations are active members of University-Industry Demonstration Partnerships (UIDP).

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